

Practical 5 - Collapsibility

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `simdata_binary.dta`, `simdata_binary2.dta`

Introduction

This practical is in three parts. In Part A we return to the kidney stones example used in session 4. Parts B and C use simulated data.

Aims

1. To explore relationships between marginal and conditional treatment effect estimates, considering both collapsible and non-collapsible quantities.
2. To gain further practice at interpreting marginal and conditional treatment effect estimates.
3. To extend the scenarios considered in the notes to include a continuous treatment-outcome confounder.

Part A: Kidney stones example

In session 4 and practical 4 we estimated the marginal and conditional treatment effects (of X on Y) using risk differences, risk ratios and odds ratios. See Tables 4.2 and 4.3.

Question 1

In the notes we saw that the marginal risk difference is equal to a weighted average of the conditional risk differences (conditional on stone size Z). What are the weights? Show that the marginal risk difference is equal to a weighted average of the conditional risk differences.

SOLUTION

It was found that the conditional risk difference in the $Z = 0$ group is 0.064 and the conditional risk difference in the $Z = 1$ group is 0.043. Using standardization we also found that the marginal risk difference is 0.054. According to the results in the notes, the marginal risk difference (RD) is a weighted average of the conditional risk differences (RD_0 and RD_1):

$$RD = RD_0 \times \Pr(Z = 0) + RD_1 \times \Pr(Z = 1)$$

In this data set we have $\Pr(Z = 0) = 0.51$ and $\Pr(Z = 1) = 0.49$. The weighted average is $0.064 \times 0.51 + 0.043 \times 0.49 = 0.054$.

Question 2

In the notes we saw that the marginal risk ratio is equal to a weighted average of the conditional risk ratios (Conditional on stone size Z). What are the weights? Show that the marginal risk ratio is equal to a weighted average of the conditional risk ratios.

SOLUTION

The marginal risk ratio is a weighted average of the conditional risk ratios given $Z = 0, 1$, with weights $w_0 = \Pr(Y = 1|do(X = 0), Z = 0) \Pr(Z = 0) / \Pr(Y = 1|do(X = 0))$ and $w_1 = \Pr(Y = 1|do(X = 0), Z = 1) \Pr(Z = 1) / \Pr(Y = 1|do(X = 0))$.

The estimated conditional risk ratios in the $Z = 0$ and $Z = 1$ groups are $RR_0 = 1.933$ and $RR_1 = 1.158$ respectively. For the weights we can estimate $\Pr(Y = 1|do(X = 0), Z = 0)$ by assuming $\Pr(Y = 1|do(X = 0), Z = 0) = \Pr(Y = 1|X = 0, Z = 0)$, i.e. that Z is the only confounder of the association between X and Y . Under the same assumption, we can use standardization to estimate $\Pr(Y = 1|do(X = 0))$. The estimates of the weights are $\hat{w}_0 = 0.210$ and $\hat{w}_1 = 0.790$. The weighted average is therefore

$$1.933 \times 0.210 + 1.158 \times 0.790 = 1.32,$$

which is the marginal risk ratio estimate, as found in practical 4.

Question 3

Can the marginal odds ratio be expressed as a weighted average of the conditional odds ratios? Explain your answer.

SOLUTION

No, because odds ratios are non-collapsible. We can show this algebraically (you were not expected to do this for this question). Consider a logistic regression model for Y given X and Z :

$$\Pr(Y|X, Z) = \frac{e^{\beta_0 + \beta_X X + \beta_Z Z}}{1 + e^{\beta_0 + \beta_X X + \beta_Z Z}}$$

with X and Z independent, so there is no confounding. Supposing that Z is binary for simplicity, to find $\Pr(Y|X)$ we can appeal to the law of total expectation (standardization), which (applied here) says that

$$\begin{aligned} \Pr(Y|X) &= \Pr(Y|X, Z = 1) \Pr(Z = 1) + \Pr(Y|X, Z = 0) \Pr(Z = 0) \\ &= \frac{e^{\beta_0 + \beta_X X + \beta_Z}}{1 + e^{\beta_0 + \beta_X X + \beta_Z}} p_Z + \frac{e^{\beta_0 + \beta_X X}}{1 + e^{\beta_0 + \beta_X X}} (1 - p_Z) \end{aligned}$$

where $p_Z = \Pr(Z = 1)$. The expression for $\Pr(Y|X)$ cannot be expressed in terms of a logistic model, i.e. we *cannot* write it in the form

$$\Pr(Y|X) = \frac{e^{\beta_0^* + \beta_X X}}{1 + e^{\beta_0^* + \beta_X X}}$$

So, after averaging/marginalizing over Z the model for $\Pr(Y|X)$ no longer takes the form of a logistic regression model.

For contrast, let's consider risk ratios estimated using a GLM with a log link. Under this model we have the conditional model

$$\Pr(Y|X, Z) = e^{\beta_0 + \beta_X X + \beta_Z Z}$$

Supposing as above that Z is binary, we have

$$\begin{aligned} \Pr(Y|X) &= e^{\beta_0 + \beta_X X + \beta_Z \times 1} \Pr(Z = 1) + e^{\beta_0 + \beta_X X + \beta_Z \times 0} \Pr(Z = 0) \\ &= e^{\beta_0 + \beta_X X + \beta_Z} p_Z + e^{\beta_0 + \beta_X X} (1 - p_Z) \\ &= e^{\beta_0 + \beta_X X + \beta_Z} p_Z + e^{\beta_0 + \beta_X X} (1 - p_Z) \\ &= e^{\beta_0 + \beta_X X} \{ \beta_Z p_Z + (1 - p_Z) \} \\ &= e^{\beta_0^* + \beta_X X} \end{aligned}$$

where $\beta_0^* = \beta_0 + \log \{ e^{\beta_Z} p_Z + (1 - p_Z) \}$. This shows that log risk ratio without adjusting for Z is β_X , the same value as in the adjusted analysis. Therefore the GLM with a log-link does possess the collapsibility property.

Discuss: What are the interpretations of the marginal and conditional risk differences, risk ratios and odds ratios.

SOLUTION

See Practical 4 solutions.

Discuss: For a patient with a large kidney stone, which treatment effect estimate(s) are most relevant to inform their treatment?

SOLUTION

For a patient with a large kidney stone, the conditional treatment effect (conditional on $Z = 1$) is most relevant. The marginal treatment effect estimate is of more relevance to a policy maker.

Part B: A simulated example with binary outcome

In this part we some simulated data (`simdata_binary.dta`) on a binary treatment X , binary outcome Y and binary covariate Z . The covariate Z is known to be measured temporally prior to X , meaning that we can be sure it does not lie on the causal pathway between X and Y .

Question 4

Load the data using `haven::read_dta` (or with use `simdata_binary.data` in Stata). Then quantify the relationship between X and Z .

SOLUTION

```
simdata_binary <- haven::read_dta("simdata_binary.dta")
table(simdata_binary$x, simdata_binary$z)
```

```
      0      1
0 1000 1000
1 1000 1000
```

There is no marginal association between X and Z . That is $\Pr(X = x|Z = 0) = \Pr(X = x|Z = 1)$.

Question 5

Using a logistic regression that includes X , Z and their interaction, estimate the conditional probabilities $\Pr(Y = 1|X = x, Z = z)$ for $x = 0, 1$ and $z = 0, 1$, and hence the conditional risk differences

$$\Pr(Y = 1|X = 1, Z = z) - \Pr(Y = 1|X = 0, Z = z), \quad z = 0, 1$$

SOLUTION

```
interMod <- glm(y~x+z+x*z, data=simdata_binary,  
               family=binomial)  
summary(interMod)
```

Call:

```
glm(formula = y ~ x + z + x * z, family = binomial, data = simdata_binary)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.197e+00	1.054e-01	-20.85	<2e-16 ***
x	2.197e+00	1.229e-01	17.88	<2e-16 ***
z	2.197e+00	1.229e-01	17.88	<2e-16 ***
x:z	1.938e-15	1.738e-01	0.00	1

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 5545.2 on 3999 degrees of freedom
Residual deviance: 4072.9 on 3996 degrees of freedom
AIC: 4080.9

Number of Fisher Scoring iterations: 4

The four fitted log odds are then given by

```
#x=0, z==0  
coef(interMod)[1]
```

```
(Intercept)  
-2.197225
```

```
#x=0, z==1
coef(interMod)[1]+coef(interMod)[3]
```

```
(Intercept)
1.332268e-15
```

```
#x=1, z==0
coef(interMod)[1]+coef(interMod)[2]
```

```
(Intercept)
1.509903e-14
```

```
#x=1, z==1
coef(interMod)[1]+coef(interMod)[3]+coef(interMod)[2]+coef(interMod)[4]
```

```
(Intercept)
2.197225
```

We can use `plogis` to convert each to a probability. The conditional risk differences required are thus

```
#risk difference at Z=0
plogis(coef(interMod)[1]+coef(interMod)[2]) - plogis(coef(interMod)[1])
```

```
(Intercept)
0.4
```

```
#risk difference at Z=1
plogis(coef(interMod)[1]+coef(interMod)[3]+coef(interMod)[2]+coef(interMod)[4]) -
plogis(coef(interMod)[1]+coef(interMod)[3])
```

```
(Intercept)
0.4
```

Thus here there is no effect modification on the risk difference scale for the effect of X on Y .

Since both covariates are categorical and we have included their interaction in the model, the model estimates are exactly equal to the corresponding sample quantities, e.g.

```
#risk difference at Z=0
mean(simdata_binary$y[(simdata_binary$x==1) & (simdata_binary$z==0)])-
  mean(simdata_binary$y[(simdata_binary$x==0) & (simdata_binary$z==0)])
```

```
[1] 0.4
```

Lastly, note that we can obtain the same results using the `marginalEffects` package with

```
library(marginalEffects)
comparisons(interMod,variables=c("x"),
            by="z")
```

z	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0	0.4	0.0184	21.7	<0.001	344.2	0.364	0.436
1	0.4	0.0184	21.7	<0.001	344.2	0.364	0.436

```
Term: x
Type: response
Comparison: 1 - 0
```

Question 6

Without performing any further calculations, what do you expect the marginal risk difference to be?

SOLUTION

Because risk differences are collapsible, and because there is no modification of the effect of X on Y by Z , we expect the marginal risk difference to be equal to the two conditional risk differences, i.e. to be 0.4.

Question 7

From your logistic regression in question 5, what are the estimates of the conditional odds ratios

$$\frac{\Pr(Y = 1|do(X = 1), Z = z)/\Pr(Y = 0|do(X = 1), Z = z)}{\Pr(Y = 1|do(X = 0), Z = z)/\Pr(Y = 0|do(X = 0), Z = z)}, \quad z = 0, 1$$

SOLUTION

These conditional OR for the effect of X among $Z = 0$ is

```
exp(coef(interMod)[2])
```

```
x  
9
```

For the conditional OR among those with $Z = 1$ we must add in (on the log odds scale) the interaction effect:

```
exp(coef(interMod)[2]+coef(interMod)[4])
```

```
x  
9
```

We obtain an identical value - this because the estimated interaction coefficient is zero.

Here manually calculating the conditional ORs is both instructive and easy. But if we did want to achieve the same using `marginalEffects`, we can use:

```
comparisons(interMod,variables=c("x"),  
             by="z",transform="exp",  
             type="link")
```

z	Estimate	Pr(> z)	S	2.5 %	97.5 %
0	9	<0.001	235.0	7.07	11.5
1	9	<0.001	235.0	7.07	11.5

```
Term: x  
Type: link  
Comparison: 1 - 0
```

Adding `type="link"` makes `comparisons` calculate the effect on the link scale (i.e. logit scale here), and then at the end apply the `"exp"` transformation to convert to an OR.

Question 8

Fit another logistic regression to obtain an estimate of the marginal odds ratio

$$\frac{\Pr(Y = 1|do(X = 1))/\Pr(Y = 0|do(X = 1))}{\Pr(Y = 1|do(X = 0))/\Pr(Y = 0|do(X = 0))}$$

What assumption do you make? Explain the difference between this estimate and your estimates in question 7.

SOLUTION

In this example Z is not confounder, as we saw there was no association between X and Z . Therefore the marginal OR above can be estimated using a logistic regression of Y on X , i.e. we do not need to use standardization to estimate the probabilities $\Pr(Y = 1|do(X = x))$. Fitting the logistic regression of Y on X , we have

```
q8Mod <- glm(y~x, data=simdata_binary,  
             family="binomial")  
exp(coef(q8Mod)[2])
```

```
      x  
5.444444
```

The marginal OR estimate is quite different from the conditional estimate (5.44 vs 9) - this is not due to confounding (there is no confounding here), but due to non-collapsibility.

Part C: A simulated example with binary outcome and continuous confounder

In this part we some simulated data (`simdata_binary2.dta`) on a binary treatment X , binary outcome Y and continuous variable Z , where Z confounds the association between X and Y .

Question 9

Fit a logistic regression of Y on X and Z and their interaction. What is the conditional odds ratio for an individual with (a) Z equal to its median value in the data, (b) Z equal to its 10th percentile in the data, (c) Z equal to its 90th percentile in the data.

SOLUTION

```
simdata_binary2 <- haven::read_dta("simdata_binary2.dta")
interMod <- glm(y~x+z+x*z, data=simdata_binary2,
               family=binomial)
summary(interMod)
```

Call:

```
glm(formula = y ~ x + z + x * z, family = binomial, data = simdata_binary2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.60630	0.49700	-9.268	< 2e-16 ***
x	1.65704	0.73257	2.262	0.0237 *
z	0.07230	0.01834	3.942	8.08e-05 ***
x:z	-0.06265	0.02535	-2.471	0.0135 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1881.1 on 3999 degrees of freedom
 Residual deviance: 1864.7 on 3996 degrees of freedom
 AIC: 1872.7

Number of Fisher Scoring iterations: 5

We can find the required quantiles of z using

```
quants <- quantile(simdata_binary2$z, probs = c(0.1, 0.5, 0.9))
quants
```

10%	50%	90%
20.03243	27.56752	35.10481

The conditional OR for the effect of X at the 10% quantile of Z is given by

```
exp(coef(interMod)[2]+20.03243*coef(interMod)[4])
```

x
1.494859

and similarly for the effect at the median and 90% quantile.

We can also calculate these using `comparisons` with:

```
comparisons(interMod, variables=c("x"),
            newdata=datagrid(z=quants),
            transform="exp",
            type="link")
```

z	Estimate	Pr(> z)	S	2.5 %	97.5 %
20.0	1.495	0.1118	3.2	0.911	2.454
27.6	0.932	0.6173	0.7	0.708	1.227
35.1	0.581	0.0137	6.2	0.378	0.895

Term: x
Type: link
Comparison: 1 - 0

Question 10

Using the empirical standardization method introduced in session 4, manually obtain an estimate of the effect of X on Y using a marginal odds ratio. Then use the `avg_comparisons` function in the `marginalEffects` package (or `margins` in Stata) to check your result.

SOLUTION

We first manually estimate the required quantities.

```
#calculate fitted probabilities under x=0, and average
prx0 <- mean(plogis(coef(interMod)[1] +
                  coef(interMod)[3]*simdata_binary2$z))
#and under x=1
prx1 <- mean(plogis(coef(interMod)[1] + coef(interMod)[2] +
                  coef(interMod)[3]*simdata_binary2$z +
                  coef(interMod)[4]*simdata_binary2$z))
```

The required marginal OR can then be calculated as

```
prx1/(1-prx1) / (prx0/(1-prx0))
```

```
[1] 0.8708034
```

Now using `avg_comparisons`, we can use

```
avg_comparisons(interMod,
                 variables="x",
                 comparison="lnor",
                 transform = "exp")
```

```
Estimate Pr(>|z|)   S 2.5 % 97.5 %
      0.871      0.346 1.5 0.653   1.16
```

Term: x

Type: response

Comparison: $\ln(\text{odds}(1) / \text{odds}(0))$

Question 11

Compare this with the odds ratio from a regression of Y on X alone. What is the reason for the difference between this estimate and that in question 10?

SOLUTION

```
noAdj <- glm(y~x,data=simdata_binary2,
             family=binomial)
exp(coef(noAdj)[2])
```

```
      x
1.088437
```

The crude marginal OR estimate from the regression of Y on X alone is 1.09. This differs from the estimate in question 10 (0.87) because the estimate in question 10 is adjusted for Z - i.e. we used standardization to adjust for Z and estimate the marginal probabilities $\Pr(Y = 1 | do(X = x))$. The crude marginal OR estimate is a biased estimate of the causal marginal OR because it does not control for confounding by Z .

Question 12

Discuss: What are the interpretations of the marginal and conditional odds ratios?

SOLUTION

The (causal) marginal OR is 0.87. The interpretation is that if everyone in this population were to receive the treatment ($X = 1$) the estimated odds of $Y = 1$ would be 13% lower than had everyone not received the treatment ($X = 0$).

The conditional OR for an individual with Z at the median value in the population ($Z = 27.57$) is estimated to be 0.93. This means that if a person with this value of Z received the treatment their estimated odds of $Y = 1$ would be 7% lower than had they not received the treatment.

The conditional OR for an individual with Z at the 10th percentile value in the population ($Z = 20.03$) is estimated to be 1.49. This means that if a person with this value of Z received the treatment their estimated odds of $Y = 1$ would be 49% higher than had they not received the treatment.

The conditional OR for an individual with Z at the 90th percentile value in the population ($Z = 35.11$) is estimated to be 0.58. This means that if a person with this value of Z received the treatment their estimated odds of $Y = 1$ would be 42% lower than had they not received the treatment.